

A Multiscale Systems Biology Model investigating the discordance between PET Imaging and CSF biomarkers in Alzheimer's Disease



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Introduction

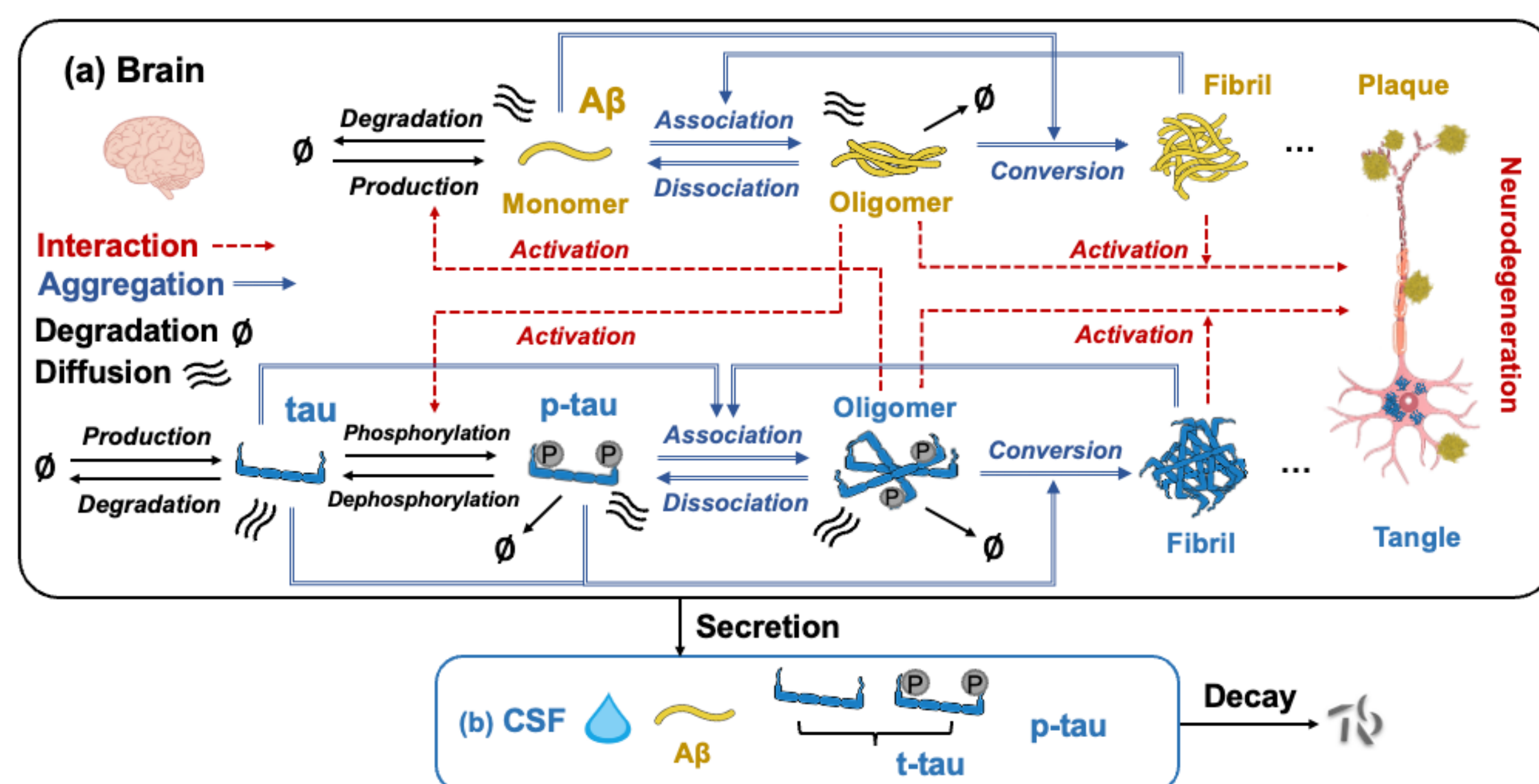
Accurate diagnosis of Alzheimer's disease (AD) relies on two categorical biomarkers: Positron Emission Tomography (PET), reflecting insoluble fibril aggregation and Cerebrospinal Fluid (CSF), measuring soluble monomer concentration. However, substantial evidence shows that imaging and fluid biomarkers yield discordant results, complicating early detection and intervention. This study develops a multiscale model to understand the neurobiological factors behind PET-CSF discrepancy. In short, our multiscale model want to

1. Simulate AD biomarkers' spread pattern with high accuracy,
2. Investigate biomarkers' discordance as a result of complex neuropathological events.

Method

AD is characterized by the accumulation of amyloid- β ($A\beta$) and tau (T), leading to amyloid plaques and tau tangles in the brain, resulting in neurodegeneration ([N]). Here, a multiscale AT[N] cascade model is proposed based on several typical AD pathways:

- Monomeric $A\beta$ and T go through natural production, while all monomers and oligomers can be degraded following the Law of Mass Action;
- $A\beta$ and tau undergo aggregation through homogeneous and fibril-catalyzed nucleation;
- multistep conversion process from soluble monomeric and oligomeric species to insoluble fibrillar species are also considered [1];
- Based on the latest neuropathological findings, we integrate the cross activations between $A\beta$ and tau in our model [2, 3].



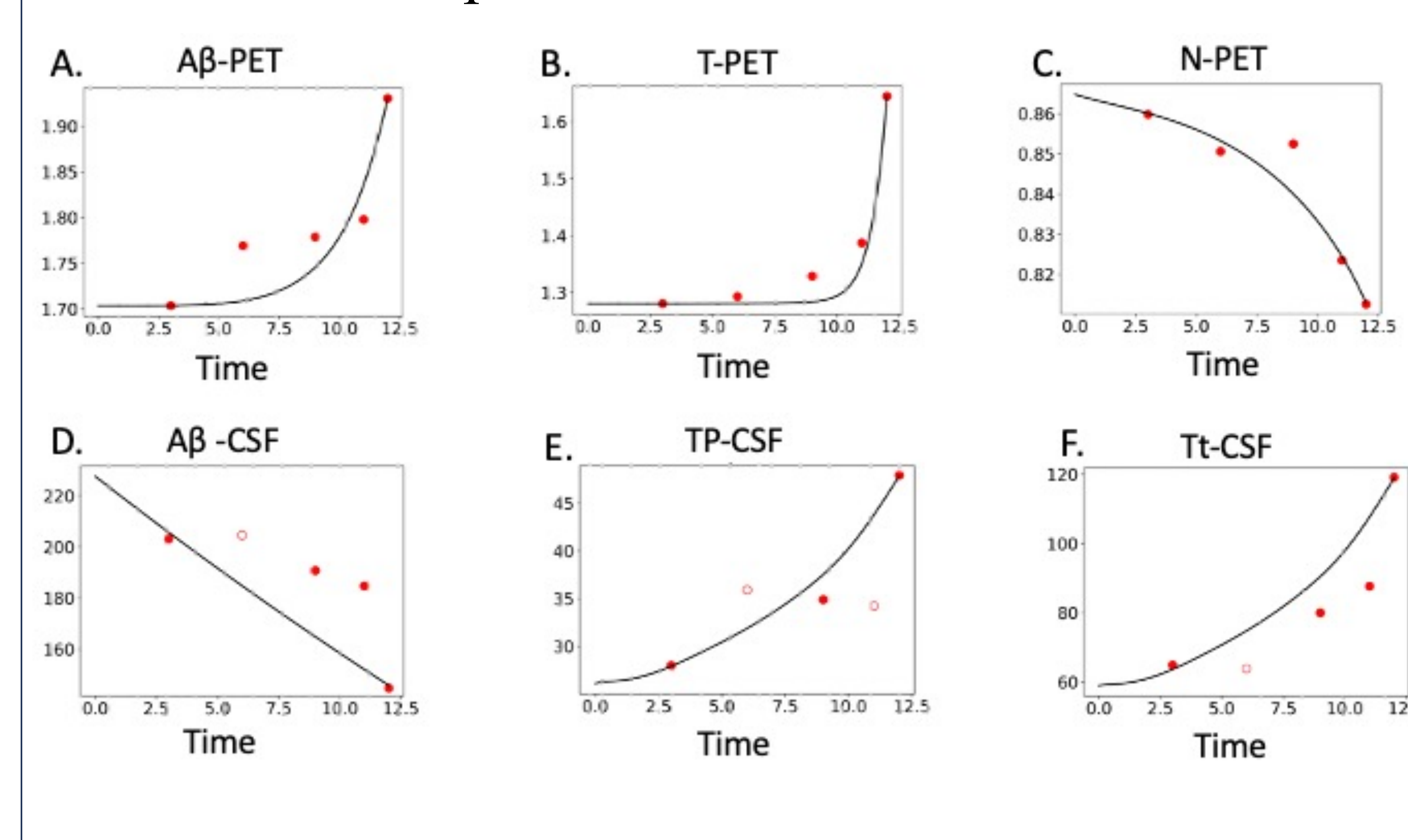
Our model involves 11 partial differential equations, simulating the kinetics of protein reactions and diffusion. We use the Hill function to approximate the complex multi-molecular interacting process in the activation pathways. The entire brain is parceled into 160 regions, comprising 148 cortical regions and 12 subcortical regions. The diffusion component of our model is constructed based on prion-like manner of white matter fiber pathways. We apply a common network method to estimate the adjacency matrix and derive the Laplacian matrix, which is used as the diffusion operator.

Reference:

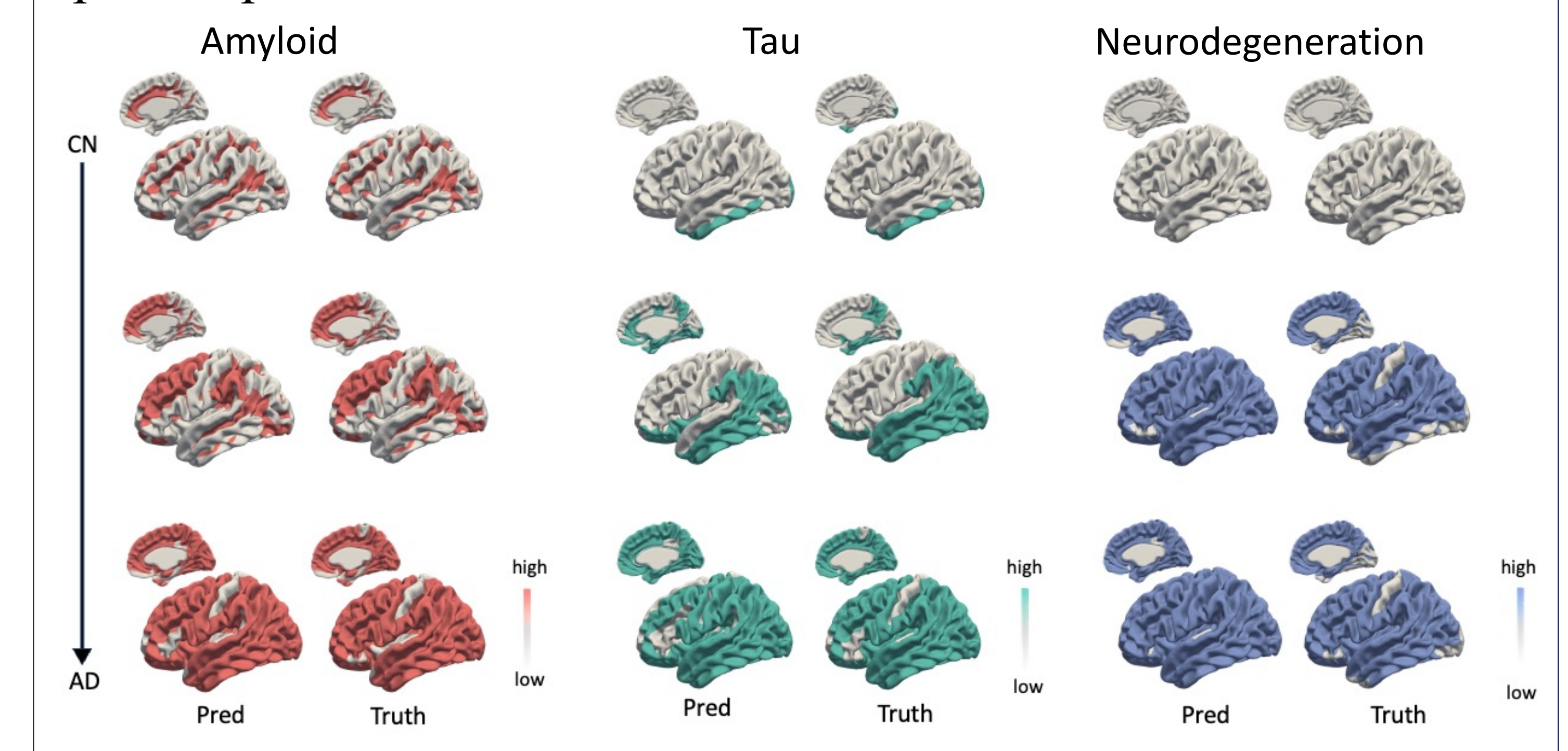
- [1] A. J. Dear, T. C. T. Michaels, G. Meisl, D. Klenerman, S. Wu, S. Perrett, S. Linse, C. M. Dobson, and T. P. J. Knowles. Kinetic diversity of amyloid oligomers. *Proceedings of the National Academy of Sciences of the United States of America*, 117(22):12087–12094, June 2020.
- [2] A. F. T. Arnsten, D. Datta, K. Del Tredici, and H. Braak. Hypothesis: Tau pathology is an initiating factor in sporadic Alzheimer's disease. *Alzheimer's & Dementia*, 17(1):115–124, Jan. 2021.
- [3] M. A. Chabrier, M. Blurton-Jones, A. A. Agazaryan, J. L. Nerhus, H. Martinez-Coria, and F. M. LaFerla. Soluble A Promotes Wild-Type Tau Pathology In Vivo. *The Journal of Neuroscience*, 32(48):17345–17350, Nov. 2012.

Result

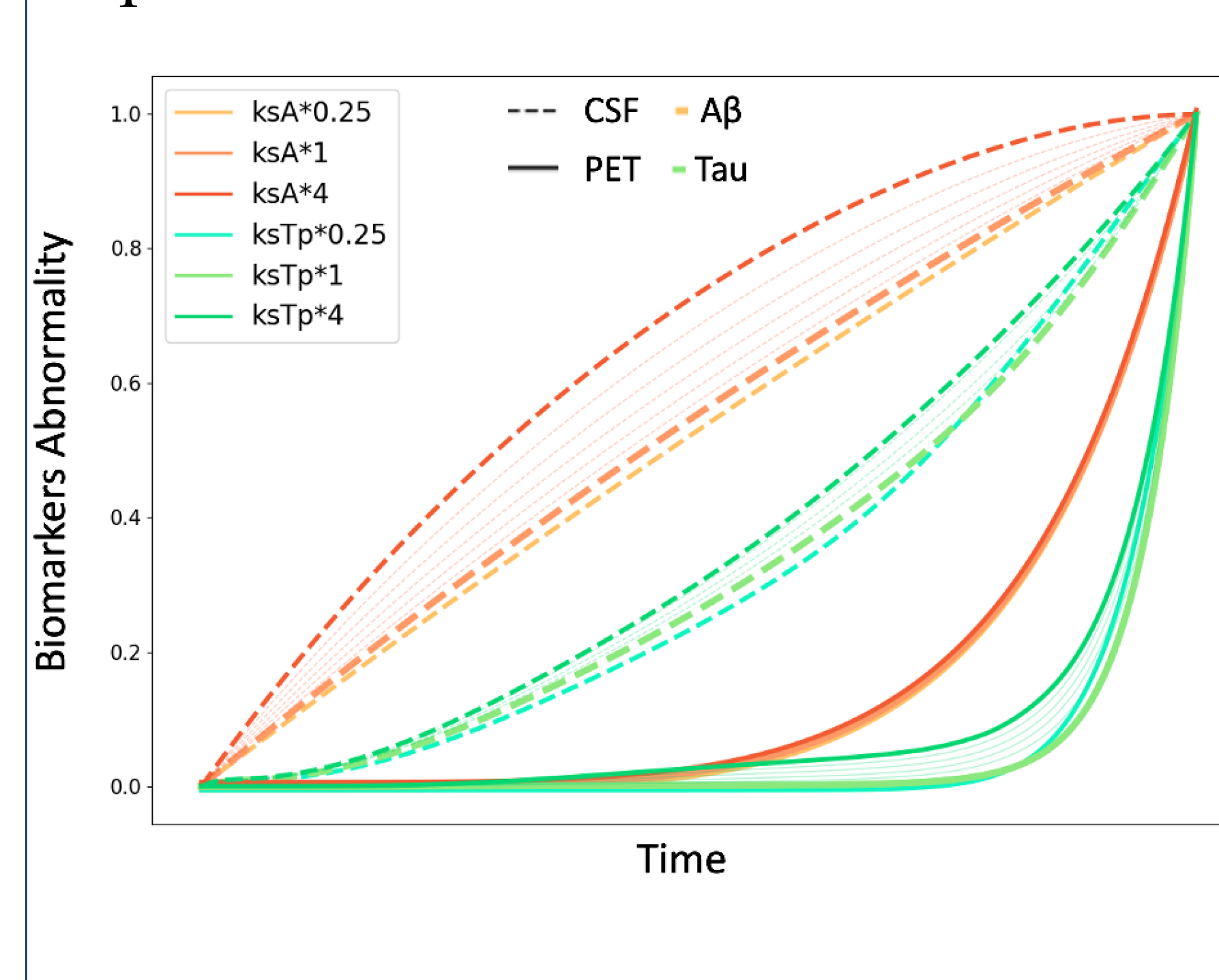
Biomarkers' temporal evolutions:



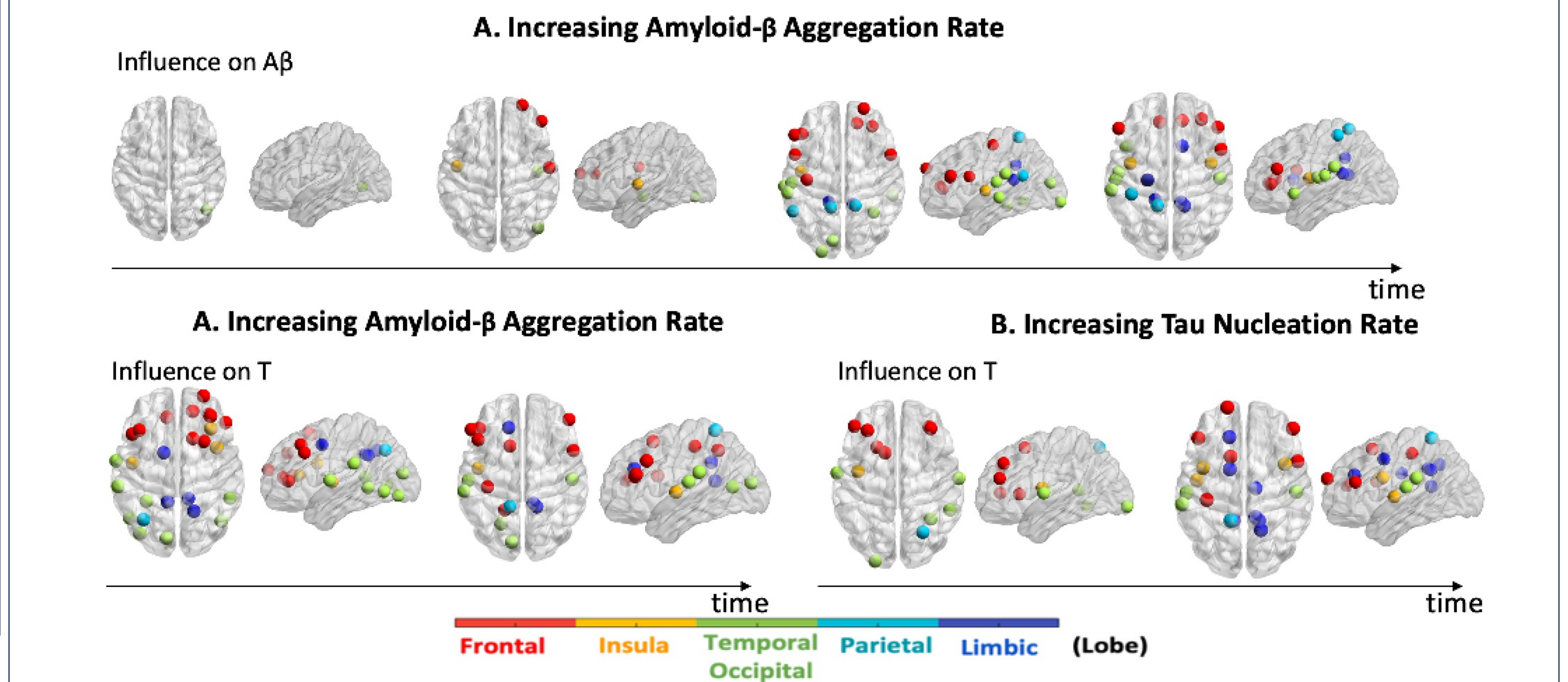
Spatial Spread of biomarkers:



Impact of Secretion Rate on CSF-PET :



Influence of $A\beta$ Aggregation/ T Nucleation



Conclusion

- The proposed multiscale model predicts the spatial pattern of $A\beta$ and tau biomarkers in different stages with high accuracy compared with imaging data.
- The results of the model demonstrates distinct $A\beta$ accumulation patterns in PET and CSF, where the soluble CSF $A\beta$ first becomes abnormal, and then the insoluble $A\beta$ accumulation is detected in PET, which is a result of integrated effects of $A\beta$ metabolism and diffusion.